To comprehend how light may be simultaneously an electromagnetic wave and a swarm of photons will require the entire construction of quantum mechanics. But the first building block of this theory has been established: there exists a fundamental granularity in *all* things, including light.

Niels, Werner and Paul

If Planck is the biological father of the theory, Einstein is the parent who gave birth to and nurtured it. But as is often the case with children, the theory then went its own way, barely recognized by Einstein as his own.

During the first two decades of the twentieth century, it is the Dane Niels Bohr who is responsible for guiding its development. Bohr studies the structure of atoms, which was beginning to be explored at the turn of the century. Experiments had shown that an atom is like a small solar system: the mass is concentrated in a heavy central nucleus, around which light electrons revolve, more or less like the planets around the Sun. This picture, however, did not account for a simple fact: matter is coloured.

Salt is white, pepper is black, chilli is red. Why? Studying the light emitted by atoms, it is apparent that substances have specific colours. Since colour is the frequency of light, light is emitted by substances at certain fixed frequencies. The set of the frequencies that characterizes a given substance is known as the ‘spectrum’ of this substance. A spectrum is a collection of fine lines of different hues, in which the light emitted by a given substance is decomposed (for instance, by a prism). The spectra of a few elements are shown in figure 4.2.



Figure 4.1 Niels Bohr.

Spectra of numerous substances had been studied and catalogued in many laboratories at the turn of the century, and nobody knew how to explain why each substance had this or that spectrum. What determines the colour of those lines?